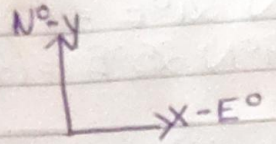
Escape the Classroom

From flight board students will have the following cities:

Alexandria	31°N, 30°E	change to (x, y)	(30, 31)
Amsterdam	52°N, 5°E	$N° \to y$	(5, 52)
Berlin	53°N, 13°N	$E° \to x$	(13, 53)
Budapest	47°N, 19°E		(19, 47)
Kyiv	50°N, 31°E		(31, 50)
Paris	49°N, 2°E		(2, 49)
Prague	50°N, 14°E		(14, 50)
Rome	42°N, 12°E		(12, 42)
Vienna	48°N, 16°E		(16, 48)



Midpoint formula $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

Alexandria to Amsterdam $\left(\frac{5+30}{2}, \frac{52+31}{2}\right) = (18, 42) = 42°N\ 18°E$
Adriatic Sea.

Alexandria to Berlin $\left(\frac{13+30}{2}, \frac{53+31}{2}\right) = (22, 42) = 42°N\ 22°E$
Kratovo

Alexandria to Budapest $\left(\frac{19+30}{2}, \frac{47+31}{2}\right) = (25, 39) = 39°N\ 25°E$
Aegean Sea

Alexandria to Kyiv $\left(\frac{31+30}{2}, \frac{50+31}{2}\right) = (31, 41) = 41°N\ 31°E$
Turkey

Alexandria to Paris $\left(\frac{2+30}{2}, \frac{49+31}{2}\right) = (16, 40) = 40°N\ 16°E$ (Italy)

Alexandria to Prague $\left(\frac{14+30}{2}, \frac{50+31}{2}\right) = (22, 40) = 40°N\ 22°E$
Thessaloniki

Alexandria to Rome $\left(\frac{12+30}{2}, \frac{42+31}{2}\right) = (21, 37) = 37°N\ 21°E$
Ionian Sea.

Alexandria to Vienna $\left(\frac{16+30}{2}, \frac{48+31}{2}\right) = (23, 39) = 40°N\ 23°E$
Achillo

From Visual Distance the closest to Athens are:

Vienna	40°N 23°E (midpoint)
Rome	37°N 21°E (midpoint)
Budapest.	39°N 25°E (midpoint)

Distance formula $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

Athens 38°N 24°E = (24, 38)
Mp Vienna 40°N 23°E = (23, 40)
Mp Rome 37°N 21°E = (21, 37)
Mp Budapest 39°N 25°E = (25, 39)

V to A $= \sqrt{(24-23)^2+(38-40)^2} = \sqrt{5}$

R to A $= \sqrt{(24-21)^2+(38-37)^2} = \sqrt{10}$

B to A $= \sqrt{(24-25)^2+(38-39)^2} = \sqrt{2}$.

The flight that stops closest to Athens is the one from Alexandria to Budapest, flight no LH0340